RoadFury: Transformer-Based Test Prioritization for Simulation-Based Testing of Self-Driving Cars

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***Abstract*—Existing test prioritization tools for self-driving car simulations reduce road geometry to a handful of aggregate statistics, discarding the sequential structure that makes a road dangerous. We present RoadFury, which preserves the full geometric sequence: a 10-channel feature matrix processed by a Pre-LN Transformer Encoder with Stochastic Weight Averaging. A systematic comparison of four paradigms—LLMs, GNNs, CNNs, and Transformers—reveals that only sequential encoding consistently outperforms random ordering. Deployed for the ICST 2026 SDC Testing Competition [7], RoadFury achieves APFD = 0.804 ± 0.012.**

***Index Terms*—test prioritization, self-driving cars, Transformer, stochastic weight averaging, road geometry**

I. Introduction

Testing self-driving cars in high-fidelity simulators like BeamNG.tech is essential but expensive [1], [2]: suites easily grow to tens of thousands of tests, each requiring tens of seconds to minutes. *Test prioritization* reorders tests so that failure-revealing cases run first, measured by APFD [3]:

$$\mathrm{APFD}(\pi) = 1 - \frac{\sum_{i=1}^{m} TF_i}{n \cdot m} + \frac{1}{2n} \tag{1}$$

where n is total tests, m failing tests, and TF_i the rank of the i-th fault (1.0: all failures first; 0.5: random). Existing tools such as SDC-Scissor [1], [2] and ITEP4SDC reduce each road to three aggregate statistics—segment length, angle change, curvature—discarding the *sequential structure* that makes a road dangerous: the precise ordering of straights followed by sudden hairpins.

This raises a natural question: *what is the right representation for road geometry?* We explored four paradigms—zero-shot LLMs [11], GNNs [16], visual CNNs [4], and sequential Transformers [8]—and found that only the last consistently surpasses random ordering. This led to **RoadFury**: a 10-channel geometric feature representation processed by a Pre-LN Transformer [8] with SWA [5]. Prior work supports these choices: Transformers outperform classical classifiers on sequential inputs [10]; early-stage ML can predict AI-artifact outcomes from structural features [9]; SWA finds wider optima [5]; Yoo and Harman [3] survey regression testing.

II. Approach

A. Road Normalization and Feature Extraction

Each test case is a variable-length sequence of 2D road control points. We resample all sequences to L=197 points via linear interpolation and extract 10 intrinsic geometric channels (Table I), yielding a (197×10) translation- and rotation-invariant feature matrix, z-normalized with training-set statistics.

TABLE I
10-Channel Road Geometry Features

Ch.	Feature	Description
f_0	Segment length	$\lVert p_{i+1} - p_i \rVert_2$
f_1	Abs. angle change	$\lvert\Delta\theta_i\rvert$
f_2	Menger curvature	$\kappa_i = 4A_i/(a_i b_i c_i)$, Heron's formula
f_3	Curvature jerk	$\Delta\kappa_i$, 1st derivative of κ
f_4	Cum. dist. norm	$\sum_{j<i} \lVert p_{j+1} - p_j \rVert / d_{\text{tot}} \in [0,1]$
f_5	Heading sin	$\sin(\theta_i)$, θ_i=atan2$(\Delta y, \Delta x)$
f_6	Heading cos	$\cos(\theta_i)$, avoids $\pm\pi$ wrap
f_7	Rel. position	$i/(L-1) \in [0,1]$
f_8	Local curv. std	std$(\kappa_{i-5:i+5})$, window = 11
f_9	Curv. accel.	$\Delta^2\kappa_i$, 2nd derivative of κ

B. RoadTransformer Architecture

The architecture (Fig. 1) reads a road as a sequence, much like a Transformer reads a sentence. Each 10-dimensional point vector is projected to 128 dimensions via a linear layer with layer normalization and GELU; a learnable `[CLS]` token is prepended and sinusoidal positional embeddings are added, yielding 198 tokens. These pass through a four-layer Pre-LN Transformer Encoder [8] (8 heads, FFN dim 512, GELU, dropout 0.1). The Pre-LN variant normalizes *before* each sub-layer, stabilizing training and avoiding gradient issues. The `[CLS]` output then passes through an MLP head (LN, 64-unit hidden layer with GELU and 20% dropout, sigmoid) to produce a failure probability $\hat{y}\in[0,1]$. The full model has ~400K parameters.

C. Training and Deployment

Training uses Adam (lr = 10^{-3}), binary cross-entropy, 75 epochs on 28,804 samples. The key ingredient is *SWA* [5]: from epoch 50, model weights are averaged over the remaining 25 epochs, finding a flatter loss basin that generalizes better to unseen road distributions. At inference, all n tests are scored in one batched forward pass and sorted by descending $\hat{y}$. The tool is containerized as a gRPC service (`Name`, `Initialize`, `Prioritize`).

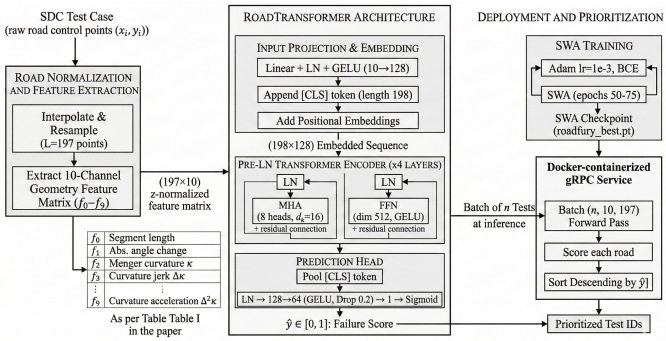

Fig. 1. RoadFury Pipeline: Road Polyline → Prioritized IDs

Fig. 1. RoadFury pipeline. Raw road control points are resampled, encoded into a 10-channel feature matrix, and processed by a Pre-LN Transformer Encoder. The `[CLS]` token summarizes the entire road; an MLP head converts it to a failure score $\hat{y}$ used for prioritization.

TABLE II
MULTI-PARADIGM COMPARISON (956 COMP. TESTS, 30 TRIALS)

Method	APFD	Paradigm
Random	0.493	–
LLM zero-shot	0.487 ± 0.019	Qwen2.5-7B, 4-bit
GNN (road graph)	0.533 ± 0.025	3-layer GCN+KNN
ResNet50 + XGBoost	0.572 ± 0.013	Road image CNN
CNN + Handcrafted	0.673 ± 0.017	ResNet50 + 44 feat.
ExtendedCNN-10ch	0.768 ± 0.014	10-ch CNN, 28K data
ITEP4SDC	0.781	ONNX MLP, 3 feat.
RoadFury (ours)	$\mathbf{0.804 \pm 0.012}$	**Transformer+SWA**

III. EVALUATION

We evaluate on SensoDat [6] (36,006 labeled BeamNG.tech tests, 80/20 split; 956-test competition set held out). All results average 30 stratified random trials.

Why not just any ML model? Table II compares four paradigms, each encoding road geometry differently.

The results reveal a clear hierarchy driven by *how much sequential structure each paradigm preserves*. LLMs score *below* random: they can discuss curvature in the abstract but cannot judge whether a specific value is dangerous. GNNs [16] capture local neighborhoods via KNN edges but discard point ordering—yet a sharp turn after a long straight is far more dangerous than the same turn after a gradual curve. Visual CNNs [4] treat roads as images, losing numeric precision; even fused with 44 handcrafted features they plateau at 0.673. The most revealing comparison is ExtendedCNN-10ch vs. pure CNN baselines: our 10-channel features in a simple 1D-CNN already reach 0.768, showing that *what* you measure matters more than *how* you model it. RoadFury's Transformer attends to all 197 points simultaneously, learning which curvature *transitions* predict failures, yielding the best APFD of 0.804.

Ablation. Seven single-factor modifications over the base Pre-LN Transformer (0.790): multi-scale conv stems, stochastic depth [15], alternative pooling, mixup [14], and TTA all *degraded* performance; focal loss [13] gave a marginal gain; only SWA [5] consistently helped (+0.014). The architecture is already well-matched to the task—regularization in weight space, not architectural complexity, is the key.

IV. CONCLUSION

How road geometry is represented matters more than model scale or training tricks. Language models, graph networks, and CNNs each discard the sequential curvature dynamics cars actually experience; a Pre-LN Transformer preserves them, achieving APFD = 0.804. Future work targets online oracle adaptation and multi-scale SWA scheduling. Code: https://github.com/chisngyen/sdc-test-prioritization-novel.

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